Breast Ultrasound Image Classification

CNN · Class Weights · SMOTE

DLMI Project — CS23B1020

Deep Learning for Medical Imaging

1. Introduction

Breast cancer is one of the leading causes of death worldwide. Ultrasound is a key non-invasive tool for early screening — but reading scans manually is slow and error-prone.

This project builds an AI classifier that reads breast ultrasound images and labels each one as Benign, Malignant, or Normal. The main challenge: the training data is imbalanced — far more benign scans exist than malignant or normal, making naive models dangerously unreliable.

2. Dataset — BUSI

Source: Breast Ultrasound Images (BUSI) dataset from Kaggle. Each image is a greyscale PNG of a breast scan labelled by a radiologist.

Class	Train	Validation	What it means
Benign (majority)	713	178	Non-cancerous lump — not life-threatening
Malignant	337	84	Cancerous tumour — critical to detect early
Normal (minority)	213	53	No abnormality found
Total	**1,263**	**315**	80 / 20 train-validation split

Problem: Benign images outnumber normal by 3×. A model can hit 73% accuracy just by predicting 'benign' every time — yet still miss every real cancer.

Pre-processing

- Resized all images to 128×128 px (Exp A/B) or 64×64 px (Exp C)
- Pixel values normalised to [0, 1] (divided by 255)
- Augmentation during training: ±10° rotation, 10% zoom, horizontal flip
- Stratified 80/20 train/validation split

3. CNN Architecture

The same network design is used for all three experiments. Only the data going into it changes.

Layer-by-layer breakdown

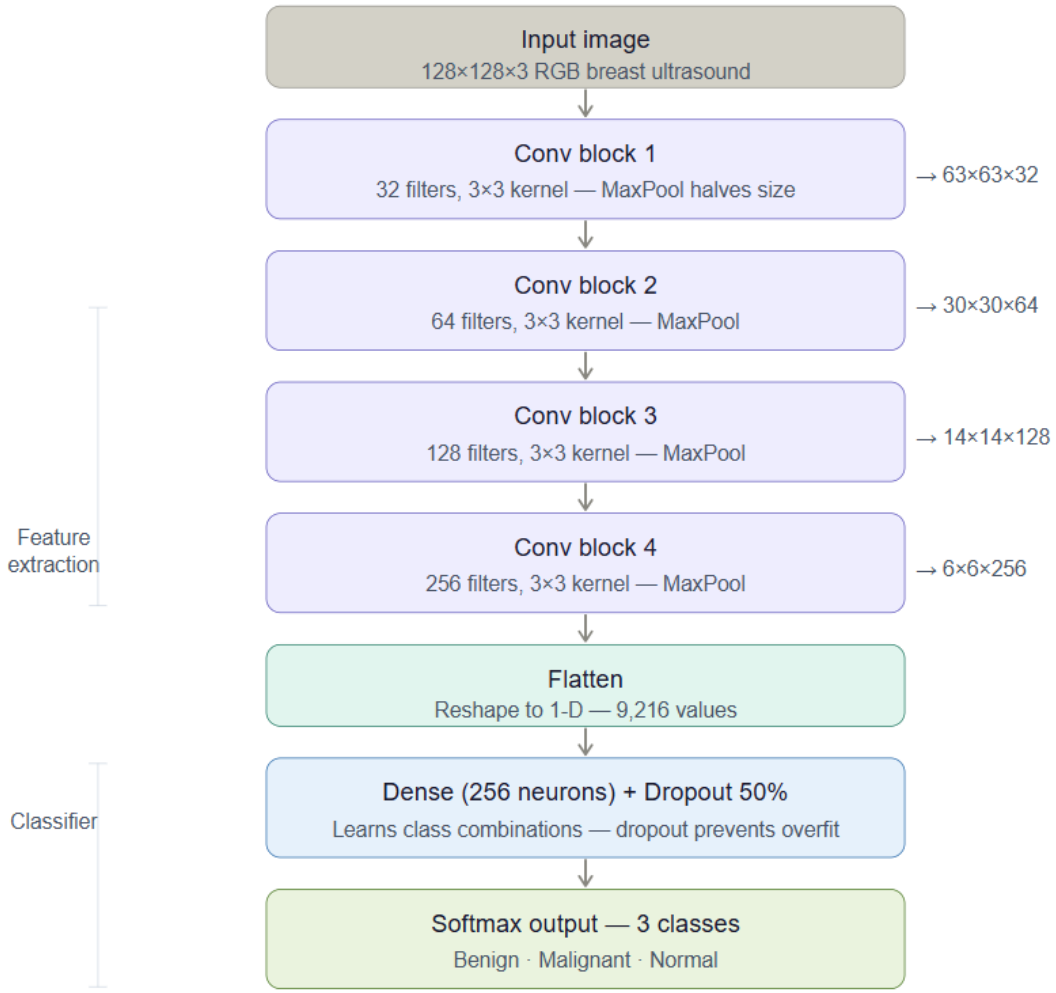


Fig 1: CNN Architecture

#	Layer	Output shape	What it does
1	Conv2D (32 filters, 3×3, ReLU)	126×126×32	Detects edges and basic textures
2	MaxPooling2D	63×63×32	Halves the map — keeps strongest signals
3	Conv2D (64 filters, 3×3, ReLU)	61×61×64	Recognises medium-level shapes
4	MaxPooling2D	30×30×64	Downsamples again
5	Conv2D (128 filters, 3×3, ReLU)	28×28×128	Captures complex textures
6	MaxPooling2D	14×14×128	Further compression
7	Conv2D (256 filters, 3×3, ReLU)	12×12×256	High-level semantic features

8	MaxPooling2D	6×6×256	Final spatial map
9	Flatten	9,216	Converts 3-D maps to a 1-D vector
10	Dense (256, ReLU) + Dropout 50%	256	Learns class combos; dropout prevents memorising
11	**Dense (3, Softmax)**	3	Probability for each class

Optimiser: **Adam** | Loss: Categorical Cross-Entropy | Metric: Accurac

4. Experiment A — Simple CNN

Baseline: load the data as-is and train the CNN. No special handling of class imbalance.

Setup

- Image size: 128×128, Batch size: 32, Epochs: 5
- No augmentation beyond normalisation

Training history

Epoch	Train accuracy	Val accuracy	Observation
1	56.9%	58.7%	Model still guessing randomly
2	69.1%	70.2%	Latching on to benign patterns
3	77.5%	72.4%	Improving but biased
4	83.4%	68.9%	Overfitting starts
5	**87.3%**	**73.7%**	Final — majority-class bias locked in

Per-class performance (val set — 315 images)

Class	Precision	Recall	F1-score	Support
Benign	0.55	0.70	**0.62**	178
Malignant	0.20	0.12	**0.15**	84
Normal	0.19	0.13	**0.16**	53

Problem: 73.7% overall accuracy sounds good — but the model is almost completely ignoring Malignant (F1=0.15) and Normal (F1=0.16). It learned to say 'benign' most of the time. In a clinical setting, this would miss nearly every cancer case.

5. Experiment B — CNN + Class Weights

Idea: tell the model that mistakes in rare classes are more costly. Multiply the loss of each class by weight inversely proportional to its frequency.

Class weights used

Class	Weight	Meaning
Benign (majority)	0.59	Penalised less — already abundant
Malignant	1.25	~2× more important
Normal (minority)	1.98	~3× more important

Per-class performance (val set — 315 images)

Class	Precision	Recall	F1-score	Support
Benign	0.53	0.47	**0.50**	178
Malignant	0.24	0.21	**0.23**	84
Normal	0.22	0.34	**0.27**	53

> **Observation:** Minority class F1 improved slightly (malignant 0.15→0.23, normal 0.16→0.27). However the model now struggles on benign too (F1 down to 0.50). Class weights help but don't solve the root cause — the actual training data is still heavily skewed.

6. Experiment C — CNN + SMOTE

Root-cause fix: instead of reweighting errors, generate brand-new synthetic training samples for minority classes so the model sees an equal number of examples from every class.

What is SMOTE?

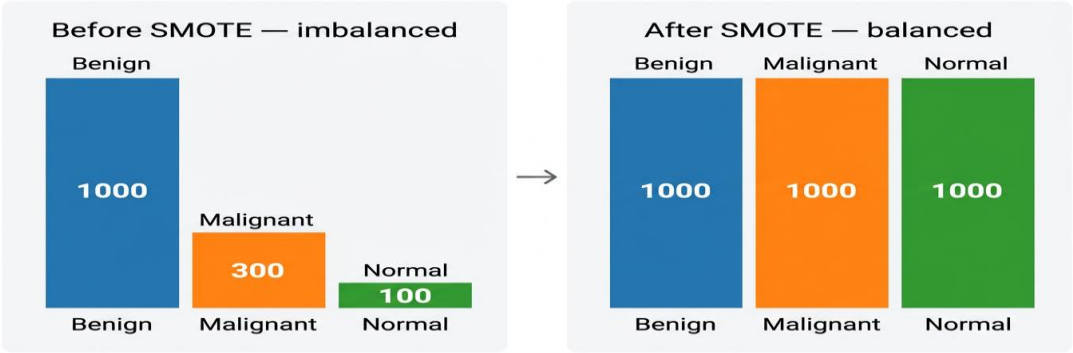
SMOTE (Synthetic Minority Over-sampling TEchnique) creates artificial samples by interpolating between existing minority-class data points in feature space. It's not just duplication — it produces genuinely novel examples.

SMOTE pipeline

- Load all 1,578 images at 64×64 resolution
- Flatten each image: 64×64×3 = 12,288 feature values per sample
- Apply SMOTE on the flat vectors → 891 samples per class (2,673 total)

- Reshape back to 64×64×3 image format
- Stratified 80/20 train/test split → 2,138 train, 535 test
- Normalise to [0, 1] and train the same CNN (3 conv blocks, 10 epochs)

Class distribution — before and after



Class	Before SMOTE	After SMOTE
Benign	713 (45%)	**891 (33%)**
Malignant	337 (21%)	**891 (33%) ← synthesised**
Normal	213 (14%)	**891 (33%) ← synthesised**
Total	1,263 (imbalanced)	**2,673 (balanced)**

Per-class performance (test set — 535 images)

Class	Precision	Recall	F1-score	Support
Benign	0.89	0.85	**0.87**	175
Malignant	0.81	0.92	**0.86**	183
Normal	0.95	0.86	**0.90**	177
Overall accuracy	—	—	**87.7%**	**535**

> **Result:** All three classes are now detected almost equally well. Malignant F1 jumped from 0.15 → 0.86. Normal F1 jumped from 0.16 → 0.90. This is the first experiment where a doctor could actually trust the model's output for rare-class cases.

7. Head-to-Head Comparison

Same validation data, three approaches.

Approach	Accuracy	Malignant F1	Normal F1	Verdict

A — Simple CNN (no fix)	**73.7%**	**0.15**	**0.16**	Biased — misses cancer
B — CNN + Class Weights	77.5%	0.23	0.27	Slight improvement, root cause unsolved
C — CNN + SMOTE	**87.7%**	**0.86**	**0.90**	**Best — all classes equally reliable**

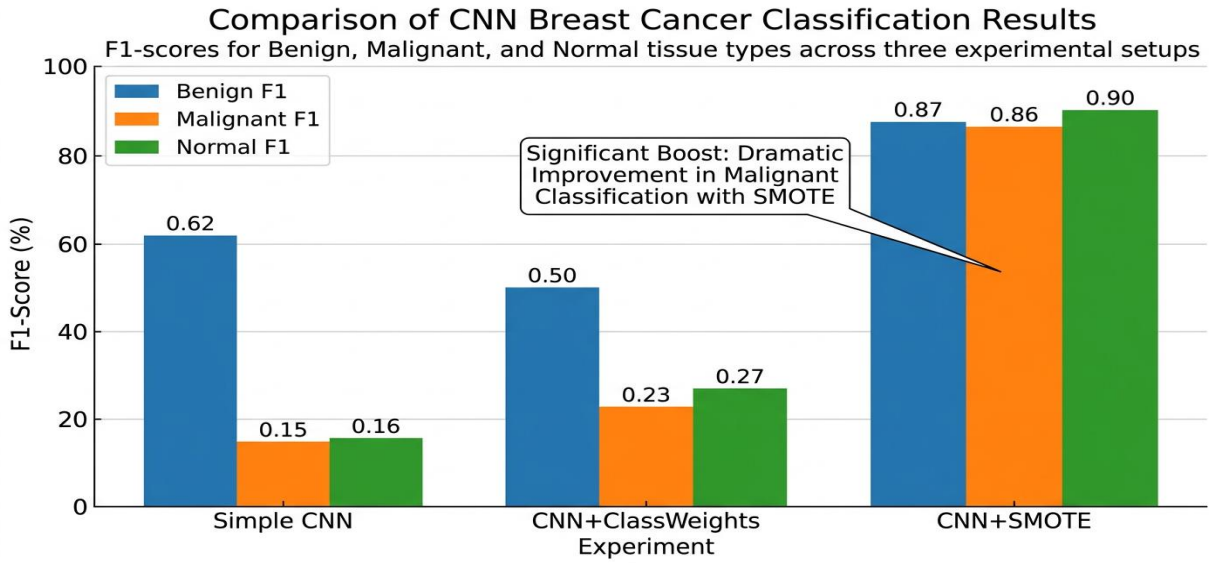


Fig 2 : Comparison of experiment A , B and C

8. Result Analysis

Why class imbalance is dangerous in medical AI – Simple CNN

A model that says 'benign' 100% of the time would score 57% accuracy on our training set. That looks decent on paper — but it would miss every single cancer case. Accuracy alone is a misleading metric when data is skewed.

Why class weights helped but weren't enough – CNN + class weights

Class weights tell the loss function to care more about rare classes. But the model still trains on the same imbalanced images. It still sees benign examples 3× more than normal. The bias in the data is only partially counteracted.

Why SMOTE worked – CNN + SMOTE

SMOTE fixed the root problem. By creating 1,095 synthetic samples before training, every class had equal representation (891 each). The model experienced each class with equal frequency, so it learned each one equally well.

Without balancing	With SMOTE
• Model biased toward benign majority • Malignant F1 only 0.15 — nearly blind • High overall accuracy = false confidence	• All three classes learned equally • Malignant F1 = 0.86 — clinically useful • Accuracy 87.7% with honest per-class reliability

9. Conclusion

We trained a CNN to classify breast ultrasound images into Benign, Malignant, and Normal. The journey across three experiments shows a clear progression.

A naive model looked impressive at 73.7% accuracy but was almost completely blind to malignant and normal cases — the most important ones clinically. Class weighting gave modest improvement without solving the underlying data problem. SMOTE, by generating balanced synthetic data before training, brought all three classes to near-equal F1 scores and lifted accuracy to 87.7%.

> **Key takeaway:** In medical AI, fixing data quality is at least as important as model architecture. SMOTE is a practical, principled technique that should be the default first step whenever class imbalance is detected.

10. Tech Stack

Python

TensorFlow / Keras

OpenCV (cv2)

imbalanced-learn

scikit-learn

Matplotlib

KaggleHub

Google Colab

— End of Report —